

Question description

Byteland has n cities and m roads between them. The goal is to construct new roads so that there is a route between any two cities. Your task is to find out the minimum number of roads required, and also determine which roads should be built.

Constraints

$$1 \leq n \leq 10^5$$

$$1 \leq m \leq 2 \cdot 10^5$$

$$1 \leq a, b \leq n$$

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After that, there are m lines describing the roads. Each line has two integers a and b : there is a road between those cities.

A road always connects two different cities, and there is at most one road between any two cities.

Output

First print an integer k : the number of required roads.

Then, print k lines that describe the new roads. You can print any valid solution.

✓ Logical Test Cases

Test Case 1
INPUT (STDIN)
4 2 1 2 3 4
EXPECTED OUTPUT
1 2 4

Test Case 2
INPUT (STDIN)
4 2 2 3 2 4
EXPECTED OUTPUT
1 1 4

✓ Mandatory Test Cases

Test Case 1
KEYWORD
<code>while(m--)</code>

✓ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
5

Test Case 2
TOKEN COUNT
475

Test Case 3
NLOC
50